Numerical Analysis I

Quiz 1A — Answer Key

Q1-01-A

An algorithm to compute x^{77} using only multiplications is

$$\begin{aligned} x_2 &= x \cdot x, \\ x_3 &= x_2 \cdot x, \\ x_5 &= x_3 \cdot x_2, \\ x_{10} &= x_5 \cdot x_5, \\ x_{15} &= x_{10} \cdot x_5, \\ x_{30} &= x_{15} \cdot x_{15}, \\ x_{60} &= x_{30} \cdot x_{30}, \\ x_{75} &= x_{60} \cdot x_{15}, \\ x_{77} &= x_{75} \cdot x_2. \end{aligned}$$

This algorithm uses $\boxed{9}$ multiplications, which is optimal.

Q1-02-A

(a)

$$(14.75)_{10} = \boxed{(1110.11)_2}.$$

(b)

$$(2.7)_{10} = (10.101100110011\ldots)_2 = \boxed{(10.1\overline{0110})_2}.$$

Q1-03-A

(a)

$$\begin{aligned} (11110.01)_2 &= 2^4 + 2^3 + 2^2 + 2^1 + 2^{-2} \\ &= 16 + 8 + 4 + 2 + \frac{1}{4} \\ &= \boxed{(30.25)_{10}}. \end{aligned}$$

(b)

$$\begin{aligned} (1.0011)_2 \times 2^5 &= (1 + 2^{-3} + 2^{-4}) \times 2^5 \\ &= 32 + 4 + 2 \\ &= \boxed{(38)_{10}}. \end{aligned}$$